School of Electrical Engineering, KAIST
EE326 Introduction to Information Theory and Coding
2019 Spring

MIDTERM

Wednesday, Apr. 17, 2019
9:00am–11:30am

NAME: ____________________

Student ID: ____________________

- Don't forget to put your name and student ID.
- Record all your solutions in this answer booklet. Only this answer booklet will be considered in the grading of your exam.
- Be sure to **show all relevant work and reasoning.** A correct answer does not guarantee full credit, and a wrong answer does not guarantee loss of credit. You should clearly but concisely indicate your reasoning.

Problem	Your score	Max score
1		5
2		15
3		10
Total		30

Problem 1

Below you have five True/False questions. If you write a correct answer, you get 1 point for each question, but if you write an incorrect answer, you get -1 point for each question. If you don't write any answer, you get 0 point. (*You don't need to write down your reasonings for T/F questions.*)

a) (1 point) For any $(X, Y, Z) \sim P_{XYZ}$, it is ALWAYS true that

$$H(XYZ) \leq H(X) + H(Y|X) + H(Z|X)$$

True or False? (Circle your answer) **True** **False**

b) (1 point) For any $(X, Y, Z) \sim P_{XYZ}$, it is ALWAYS true that

$$I(X;Y) \geq I(X;Y|Z).$$

True or False? (Circle your answer) **True** **False**

c) (1 point) For any random variable $X \in \mathcal{X}$ with a finite $|\mathcal{X}|$ and any function $g : \mathcal{X} \to \mathbb{R}$, it is ALWAYS true that

$$H(g(X)) \leq H(X).$$

True or False? (Circle your answer) **True** **False**

d) (1 point) For any random variable $(X, Y) \sim P_{XY}$ where $X \in \mathcal{X}$ for a finite $|\mathcal{X}|$ and any function $g : \mathcal{X} \to \mathbb{R}$, it is ALWAYS true that

$$I(g(X);Y) \geq I(X;Y).$$

True or False? (Circle your answer) **True** **False**

e) (1 point) For any $(X, Y, Z) \sim P_{XYZ}$, it is ALWAYS true that

$$D(P_{XYZ} \| [P_X \cdot P_Y \cdot P_Z]) = I(X;Y) + I(XY;Z).$$

True or False? (Circle your answer) **True** **False**

Problem 2

In this problem, we consider source coding for i.i.d. random variables $(X_1, X_2, \ldots, X_n)$ following the Bernoulli(p) distribution, i.e., $\Pr(X_i = 1) = p$ and $\Pr(X_i = 0) = 1 - p$ for $p \in (0, 1/2)$.

Different from the formulations in class, we now allow some errors in the reconstruction. Let the length-n source sequence be $x^n = (x_1, x_2, \ldots, x_n) \in \{0,1\}^n$ and the reconstruction be $\hat{x}^n = (\hat{x}_1, \hat{x}_2, \ldots, \hat{x}_n) \in \{0,1\}^n$. Define the Hamming distance $d(x^n, \hat{x}^n)$ between two sequences as

$$d(x^n, \hat{x}^n) := \sum_{i=1}^{n} \mathbb{1}(\hat{x}_i \neq x_i). \tag{1}$$

The Hamming distance is the number of positions in $i \in \{1, 2, \ldots, n\}$ where the bit values of the two sequences $x^n, \hat{x}^n$ are different. We say a reconstruction error occurs only when

$$d(x^n, \hat{x}^n) > n\alpha. \tag{2}$$

Assume that α is a constant satisfying $0 < \alpha < p < 1/2$.

A valid source coding should make that the probability of error, as defined above, go to 0 for large enough n. We would like to see how the relaxed error condition (2), parameterized by α, allows the required source-coding rate (bits/symbol) to be decreased.

Consider the following two approaches, both of which make the probability of error go to 0 as $n \to \infty$.

<u>Approach 1</u>: The encoder uniformly and randomly picks $n\alpha$ positions where x^n has value 1, and changes them to 0's. (If there are not that many 1's, it claims an error.) Let the resulting sequence be $\hat{x}^n$. Now a lossless source code for Bernoulli($p - \alpha$) source is used to encode $\hat{x}^n$, i.e., all the typical sequences with respect to Bernoulli($p - \alpha$) are encoded by one-to-one mapping, and the other sequences are claimed as error. The decoder simply reconstruct $\hat{x}^n$, when it is the typical sequence of Bernoulli($p - \alpha$) distribution.

<u>Approach 2</u>: The encoder takes the first $n\alpha/p$ symbols of x^n (compute the average number of 1's in this subsequence), and set them to 0. The rest $n(1-\alpha/p)$ symbols are unchanged. The resulting sequence is denoted $\hat{x}^n$. Now a lossless source code for Bernoulli(p) source is used to encode only the $n(1 - \alpha/p)$ symbols, which were not changed, so the decoder can reconstruct these symbols. The sequence $\hat{x}^n$ is reconstructed simply by attaching $n\alpha/p$ 0's in front.

a) (5 points) Compute the source-coding rates of the two approaches.

Answer:
Source coding rate R_1 of Approach 1: $R_1 =$
Source coding rate R_2 of Approach 2: $R_2 =$

Reasoning for Problem 2(a):

b) (5 points) Which one achieves a lower rate? (Hint: Use concavity of entropy)

Answer: Circle your answer $R_1 \leq R_2$ $R_1 \geq R_2$

Reasoning for Problem 2(b):

c) (5 points) Come up with intuitive explanations on why Approach 1 or 2, or both, are not the most efficient source coding for the relaxed error condition.

Answer:

Problem 3

In this problem, we start considering the simplest version of a communication system. Assume that we want to send one of two messages (e.g., 'today is sunny' or 'today is not sunny') by using a noisy binary-input binary-output channel. This channel takes a binary input $X_i \in \{0,1\}$ at each time $i = 1, 2, \ldots, n$ and generates a binary output $Y_i \in \{0,1\}$, with a conditional distribution

$$\Pr(Y_i = y | X_i = x) = \begin{cases} 1-\epsilon, & \text{if } x = y \\ \epsilon, & \text{if } x \neq y. \end{cases} \tag{3}$$

for some constant $\epsilon < 1/2$. We further assume that the output Y_i at time i depends only on the input X_i but not on any other X_j's, $j \neq i$.

In order to send one of the two messages reliably (with an arbitrary small error probability) over this noisy channel, we use the following scheme. If 'today is sunny', we send the bit 0 total n times, i.e., $X_1X_2 \ldots X_n = 00 \ldots 0$, and otherwise we send the bit 1 total n times, i.e., $X_1X_2 \ldots X_n = 11 \ldots 1$. A receiver, who only observes the n channel outputs $Y_1, \ldots, Y_n$, decides that 'today is sunny' if the number of received 0's in $Y_1, \ldots, Y_n$ is larger than $n/2$ and it decides that 'today is not sunny' if the number of received 0's in $Y_1, \ldots, Y_n$ is less than or equal to $n/2$.

We now analyze the error probability in this communication system.

a) (5 points) When the true message was 'today is sunny' and thus $X_1X_2 \ldots X_n = 00 \ldots 0$, the probability that the receiver receives some sequence $y_1y_2 \ldots y_n$ with its empirical distribution $\hat{P}_{y_1^n} = [\frac{n-\alpha n}{n}, \frac{\alpha n}{n}]$ for some integer αn with an $\alpha \in [0,1]$ is

$$\Pr\left(\left\{y^n : \hat{P}_{y^n} = \left[\frac{n-\alpha n}{n}, \frac{\alpha n}{n}\right]\right\} \Big| X_1X_2 \ldots X_n = 00 \ldots 0\right) = \binom{n}{\alpha n}\epsilon^{\alpha n}(1-\epsilon)^{n-\alpha n}.$$

The communication error occurs if the receiver receives some sequence $y_1y_2 \ldots y_n$ with its empirical distribution $[\frac{n-\alpha n}{n}, \frac{\alpha n}{n}]$ for some integer αn with any $\alpha \in [1/2, 1]$. Thus, the error probability is equal to

$$P_e = \sum_{\alpha n = \text{integer},\ \alpha \in [1/2,1]} \binom{n}{\alpha n}\epsilon^{\alpha n}(1-\epsilon)^{n-\alpha n}.$$

By using 'Method of Types', calculate this error probability (up to exponential approximation) in terms of the KL divergence between two Bernoulli distributions.

$$P_e \doteq e^{-nD(\text{Bern}(\beta_1)\|\text{Bern}(\beta_2))}$$

Find the values for β_1 and β_2.

Answer:

$\beta_1 =$ $\beta_2 =$

Reasoning for Problem 3(a):

b) (5 points) In this problem, we generalize the previous communication scheme in order to allow a transmission of 4 different messages (e.g., sunny, rainy, windy, cloudy). Let us use four different channel inputs

$$X_1X_2\ldots X_n = [\underbrace{00\ldots00}_{n}], [\underbrace{00\ldots0}_{n/2}\underbrace{11\ldots1}_{n/2}], [\underbrace{11\ldots1}_{n/2}\underbrace{00\ldots0}_{n/2}], [\underbrace{11\ldots11}_{n}]$$

for each of the messages. For example, when 'today is sunny,' we transmit $X_1X_2\ldots X_n = [\underbrace{00\ldots00}_{n}]$ and when 'today is rainy,' we transmit $X_1X_2\ldots X_n = [\underbrace{00\ldots0}_{n/2}\underbrace{11\ldots1}_{n/2}]$.

Given the received channel outputs $y_1y_2\ldots y_n$ when we denote the empirical distribution of the first half of the symbols by $\hat{P}_{y_1^{n/2}} \in [\frac{n/2-\alpha_1(n/2)}{n/2}, \frac{\alpha_1(n/2)}{n/2}]$ and that of the next half by $\hat{P}_{y_{n/2}^n} \in [\frac{n/2-\alpha_2(n/2)}{n/2}, \frac{\alpha_2(n/2)}{n/2}]$ for some $\alpha_1, \alpha_2 \in [0,1]$, the receiver decides that the transmitted message is

- 'sunny' if $\alpha_1 \in [0, 1/2)$ AND $\alpha_2 \in [0, 1/2)$;
- 'rainy' if $\alpha_1 \in [0, 1/2)$ AND $\alpha_2 \in [1/2, 1]$;
- 'windy' if $\alpha_1 \in [1/2, 1]$ AND $\alpha_2 \in [0, 1/2)$;
- 'cloudy' if $\alpha_1 \in [1/2, 1]$ AND $\alpha_2 \in [1/2, 1]$.

When the true message is 'sunny,' calculate the error probability P_e (the probability that the receiver claims 'rainy' OR 'windy' OR 'cloudy') up to exponential approximation.

Answer:
$P_e \doteq$

Reasoning for Problem 3(b):